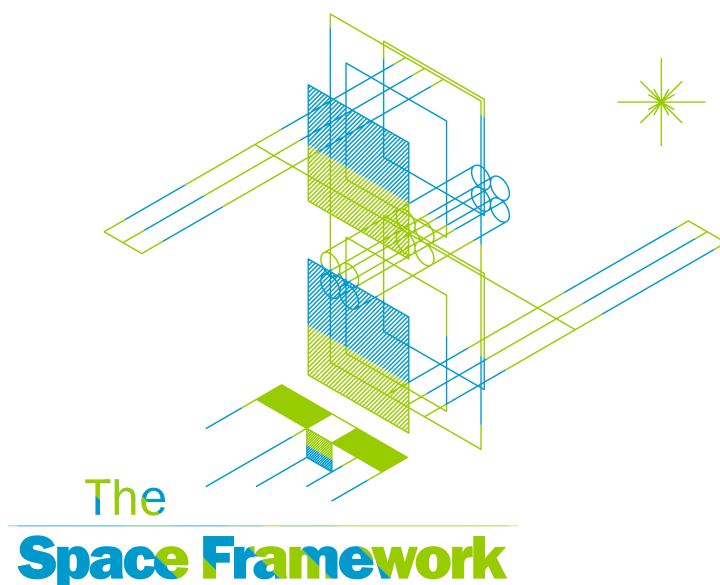


Water Electrolysis for Skyraker™ ISRU Technical Demo

Complete Theoretical, Electrochemical and Practical Guide

by Giovanni Nicola D'Aloisio



Case Study Materials (Amazon / IKEA)

Power: 36× Amazon Basics AA 1.5 V alkaline – configuration 4S × 9P (≈ 5.8 V, 26 Ah) [Amazon link](#)

Electrodes: Faber-Castell TK 9071 2 mm B leads [Amazon link](#)

Electrolyte: Anhydrous Na₂SO₄ Inoxia 1 kg [Amazon link](#)

Solvent: Demineralised water 5 L [Amazon link](#)

Collection: ONFARMA syringes 60 mL (pharmacy, $\approx$ €1 each; need 2) [Amazon link](#)

Container: IKEA PRUTA 1.8 L (23×16×8 cm) [IKEA link](#)

Indicator: Bromothymol blue (BTB) – Pharmacy or school supplier

pH strips: 1–14 universal strips (×240) [Amazon link](#)

March 3rd, 2026

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Fact-checked: NIST-JANAF, Atkins *Physical Chemistry* 11th ed., ISO 15916:2015, ISO 22734:2019

Part of the Space Framework™.



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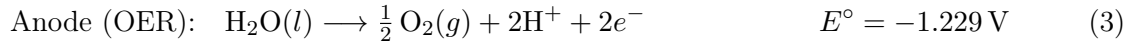
1 Overview and Global Reaction

Water electrolysis is the electrochemical decomposition of liquid water into hydrogen and oxygen, driven by an applied electric potential. The net stoichiometric reaction is:



1.1 Half-Reactions in Different Media

In **acidic medium**:



In **alkaline medium**:



In **neutral medium** (Na_2SO_4 , $\text{pH} \approx 7$) – our case study:



All standard potentials referenced to the Standard Hydrogen Electrode (SHE) at $T = 298.15 \text{ K}$, $p = 101\,325 \text{ Pa}$, unit activities [6, 7].

Why Na_2SO_4 and not NaCl ? The sulphate ion has an oxidation potential of $E^\circ = +2.01 \text{ V}$ vs SHE, much higher than water oxidation at pH 7 (+0.816 V): the electrode reaction is therefore *exclusively* the OER. By contrast, Cl^- oxidises at +1.36 V, a potential that is easily reached in a laboratory setup, producing toxic Cl_2 gas. **Never use NaCl as electrolyte.**

2 Thermodynamics of the Electrolytic Cell

2.1 Gibbs–Helmholtz Decomposition

The total enthalpy change ΔH required to drive reaction (1) decomposes into electrical work (Gibbs free energy ΔG) and a reversible heat exchange ($T\Delta S$):

$$\Delta H = \Delta G + T\Delta S \quad (8)$$

Physically: ΔG is the minimum electrical work that must be supplied; $T\Delta S$ is the heat that the cell can absorb from (or must reject to) its surroundings at constant temperature. If the cell operates below the thermoneutral voltage it must absorb heat; above it, it rejects heat.

Standard values from NIST-JANAF at $T^\circ = 298.15 \text{ K}$, $p^\circ = 101\,325 \text{ Pa}$:

$$\Delta H^\circ = 285.835 \frac{\text{kJ}}{\text{mol}} \quad (\text{Higher Heating Value, HHV}) \quad (9)$$

$$\Delta G^\circ = 237.141 \frac{\text{kJ}}{\text{mol}} \quad (10)$$

$$T\Delta S^\circ = \Delta H^\circ - \Delta G^\circ = 48.694 \frac{\text{kJ}}{\text{mol}} \quad (11)$$

2.2 Reversible Cell Voltage

The minimum electrical work per mole of H_2 is ΔG° . With $z = 2$ electrons transferred per molecule of H_2 and the Faraday constant $F = 96\,485.332 \frac{\text{C}}{\text{mol}}$:

$$V_{\text{rev}} = \frac{\Delta G}{zF} \quad (12)$$

$$V_{\text{rev}}^\circ = \frac{237\,141 \frac{\text{J}}{\text{mol}}}{2 \times 96\,485.332 \frac{\text{C}}{\text{mol}}} = 1.2289 \text{ V} \approx 1.23 \text{ V} \quad (13)$$

Below V_{rev} , the reaction cannot proceed spontaneously. Above it, water splits – but only if overpotentials (Section 3) are also overcome.

2.3 Thermoneutral Cell Voltage

$$V_{\text{th}} = \frac{\Delta H}{zF} \quad (14)$$

$$V_{\text{th}}^\circ = \frac{285\,835 \frac{\text{J}}{\text{mol}}}{2 \times 96\,485.332 \frac{\text{C}}{\text{mol}}} = 1.4812 \text{ V} \approx 1.48 \text{ V} \quad (15)$$

When $V_{\text{cell}} = V_{\text{th}}$, the electrical energy input equals ΔH : the cell operates isothermally. Below V_{th} , the cell is endothermic (SOEC regime only). Above V_{th} , excess energy is dissipated as heat.

2.4 Nernst Equation: Pressure and Temperature Corrections

For non-standard conditions:

$$V_{\text{rev}}(T, p) = V_{\text{rev}}(T) + \frac{RT}{zF} \ln \left(\frac{(p_{\text{H}_2}/p^\circ) (p_{\text{O}_2}/p^\circ)^{1/2}}{a_{\text{H}_2\text{O}}} \right) \quad (16)$$

For dilute aqueous electrolyte, $a_{\text{H}_2\text{O}} \approx 1$.

Temperature dependence (linearised):

$$V_{\text{rev}}(T) \approx V_{\text{rev}}^\circ - \frac{\Delta S^\circ}{zF} (T - T^\circ), \quad \frac{dV_{\text{rev}}}{dT} \approx -8.5 \times 10^{-4} \frac{\text{V}}{\text{K}} \quad (17)$$

Higher temperature lowers the minimum electrical work requirement – this is exploited in SOEC.

pH correction (our case, pH 7): Applying -0.05916 V/pH at 25°C :

$$E(\text{H}_2|\text{H}^+, \text{pH } 7) = 0.000 - 0.059 \times 7 = -0.414 \text{ V}$$

$$E(\text{O}_2|\text{H}_2\text{O}, \text{pH } 7) = 1.229 - 0.059 \times 7 = +0.816 \text{ V}$$

$$V_{\text{cell, min}}(\text{pH } 7) = 0.816 - (-0.414) = 1.23 \text{ V}$$

The Nernst correction cancels in the difference for this symmetric reaction: the minimum cell voltage is 1.23 V regardless of pH.

2.5 Effect of Temperature on Thermodynamic Potentials

Table 1: Thermodynamic potentials at representative operating temperatures (NIST-JANAF, $p = 1$ bar, liquid water below 100°C).

Technology	T [$^\circ\text{C}$]	V_{rev} [V]	V_{th} [V]
AWE / PEM (standard)	25	1.2289	1.4812
AWE	80	1.1840	1.4734
PEM	80	1.1840	1.4734
SOEC	800	0.9146	1.2854

At 800°C the gap between V_{rev} and V_{th} narrows, meaning a larger fraction of the energy can be supplied as cheap industrial heat rather than costly electricity.

3 Electrochemical Kinetics and Overpotentials

3.1 Real Cell Voltage Model

In practice, V_{cell} must exceed V_{rev} to drive current. The excess is partitioned into three loss mechanisms:

$$V_{\text{cell}} = V_{\text{rev}} + \eta_{\text{act}} + \eta_{\text{ohm}} + \eta_{\text{conc}} \quad (18)$$

3.2 Activation Overpotential: Butler–Volmer Equation

The activation overpotential η_{act} arises from the energy barrier that charge carriers must overcome at the electrode–electrolyte interface. The Butler–Volmer (BV) equation gives the current density j [$\frac{\text{A}}{\text{cm}^2}$] as a function of η_{act} :

$$j = j_0 \left[\exp\left(\frac{\alpha_a F \eta_{\text{act}}}{RT}\right) - \exp\left(-\frac{\alpha_c F \eta_{\text{act}}}{RT}\right) \right] \quad (19)$$

where j_0 is the *exchange current density* (a measure of intrinsic electrode kinetics), and α_a , α_c are the anodic and cathodic transfer coefficients (typically $\alpha_a + \alpha_c = 1$; for symmetric barriers $\alpha_a = \alpha_c = 0.5$).

For large overpotentials $|\eta_{\text{act}}| \gg RT/F$ (the **Tafel regime**, practical for most electrolysis conditions), eq. (19) reduces to:

$$\eta_{\text{act}} = \frac{RT}{\alpha F} \ln\left(\frac{j}{j_0}\right) = b \log_{10}\left(\frac{j}{j_0}\right) \quad (20)$$

where the Tafel slope is:

$$b = \frac{2.303 RT}{\alpha F} \approx 120 \text{ mV/decade} \quad (\alpha = 0.5, T = 25^\circ\text{C}) \quad (21)$$

Representative exchange current densities at 80°C :

- HER on Pt (cathode): $j_0 \approx 1 \times 10^{-3} \frac{\text{A}}{\text{cm}^2}$
- OER on IrO_2 (anode): $j_0 \approx 1 \times 10^{-9} \frac{\text{A}}{\text{cm}^2}$
- HER/OER on graphite (our case): $j_0 \ll 1 \times 10^{-3} \frac{\text{A}}{\text{cm}^2}$ (higher η_{act} , lower performance)

The OER has orders-of-magnitude lower j_0 than the HER: the anode is always the kinetically limiting electrode.

3.3 Ohmic Overpotential

Ionic resistance of the electrolyte and electronic resistance of current-carrying components cause a linear voltage drop:

$$\eta_{\text{ohm}} = I \cdot R_{\text{eq}} = j \cdot \text{ASR} \quad (22)$$

where R_{eq} [Ω] is the total circuit resistance and ASR [$\Omega \text{ cm}^2$] is the area-specific resistance. For reference, Nafion 117 at 80 °C: $\text{ASR} \approx 0.10 \Omega \text{ cm}^2$.

The ionic resistance of the liquid electrolyte between planar electrodes:

$$R_{\text{eq}} = \frac{1}{\kappa} \frac{d}{A} \quad (23)$$

where κ [$\frac{\text{S}}{\text{cm}}$] is the electrolyte conductivity, d [cm] is the electrode separation, and A [cm^2] is the electrode active area. This is the dominant resistance in our didactic setup (see Section 13).

3.4 Concentration Overpotential

At high current densities, mass-transport limitations cause reactant depletion near the electrode surface:

$$\eta_{\text{conc}} = \frac{RT}{zF} \ln \left(1 - \frac{j}{j_L} \right) \quad (24)$$

where j_L is the limiting current density. For our laboratory setup ($j \approx 140 \frac{\text{mA}}{\text{cm}^2} \ll j_L$), this term is negligible.

4 Electrolysis Technologies

4.1 Alkaline Water Electrolysis (AWE)

The most industrially mature technology (>100 years of deployment). Uses concentrated KOH (20–30 wt%) at 60–90 °C, with a microporous Zirfon[®] diaphragm separating the electrode compartments. Electrodes are typically Ni-based. Typical parameters: $j = 0.2\text{--}0.5 \frac{\text{A}}{\text{cm}^2}$, $V_{\text{cell}} = 1.8\text{--}2.4 \text{ V}$, specific energy consumption 50–60 kWh kg^{−1}.

4.2 Proton Exchange Membrane (PEM)

Uses a solid sulfonated polymer membrane (e.g. Nafion[®]) as both electrolyte and separator. Advantages: compact design, high current densities ($1\text{--}3 \frac{\text{A}}{\text{cm}^2}$), very high purity (> 99.999%), fast dynamic response (well-suited for coupling to intermittent renewables). Primary cost challenge: acid-stable catalysts (Ir, Pt) and membrane cost. $V_{\text{cell}} = 1.7\text{--}2.2 \text{ V}$.

4.3 Solid Oxide Electrolysis Cell (SOEC)

Operates at 700–850 °C. At these temperatures ΔG decreases and $T\Delta S$ increases (see Table 1), so a larger fraction of the total energy can be supplied as heat rather than electricity. Electrical consumption $\approx 35\text{--}42 \text{ kWh kg}^{-1}$ – the lowest of all three technologies. Key challenge: material degradation under thermal cycling and high-temperature corrosion.

5 Faraday's Laws and Gas Production Rates

5.1 Molar Production Rate (Faraday's First Law)

The total charge passed in time Δt at constant current I :

$$Q = I \Delta t \quad (25)$$

The number of moles of H_2 produced ($z = 2$ electrons per molecule):

$$n_{\text{H}_2} = \frac{I \Delta t}{z F} = \frac{I \Delta t}{2F} \quad (26)$$

By stoichiometry of eq. (1):

$$n_{\text{O}_2} = \frac{n_{\text{H}_2}}{2} = \frac{I \Delta t}{4F} \quad (27)$$

5.2 Volumetric Flow Rate (Ideal Gas)

At laboratory temperature T and pressure p (ideal gas):

$$V_{\text{H}_2} = \frac{n_{\text{H}_2} RT}{p} = \frac{I \Delta t RT}{2Fp} \quad (28)$$

Instantaneous volumetric flow rate (time derivative):

$$\dot{V}_{\text{H}_2} = \frac{IRT}{2Fp}, \quad \dot{V}_{\text{O}_2} = \frac{1}{2} \dot{V}_{\text{H}_2} \quad (29)$$

5.3 Mass Flow Rate

$$\dot{m}_{\text{H}_2} = \frac{I M_{\text{H}_2}}{2F}, \quad M_{\text{H}_2} = 2.01588 \times 10^{-3} \frac{\text{kg}}{\text{mol}} \quad (30)$$

5.4 Numerical Example at $I = 2.00 \text{ A}$ (Reference Setup)

At $T = 293.15 \text{ K}$ (20°C), $p = 101325 \text{ Pa}$:

$$\dot{V}_{\text{H}_2} = \frac{2.00 \times 8.31446 \times 293.15}{2 \times 96485 \times 101325} = 2.49 \times 10^{-7} \frac{\text{m}^3}{\text{s}} = 0.249 \frac{\text{mL}}{\text{s}} \quad (31)$$

$$\dot{V}_{\text{O}_2} = 0.124 \frac{\text{mL}}{\text{s}} \quad (32)$$

$$\dot{m}_{\text{H}_2} = \frac{2.00 \times 2.016 \times 10^{-3}}{2 \times 96485} = 2.089 \times 10^{-8} \frac{\text{kg}}{\text{s}} = 20.89 \frac{\mu\text{g}}{\text{s}} \quad (33)$$

In $\Delta t = 600 \text{ s}$ (10 min): $V_{\text{H}_2} \approx 149 \text{ mL}$, $V_{\text{O}_2} \approx 74.5 \text{ mL}$.

6 Efficiency Metrics

6.1 Faradaic Efficiency (Current Efficiency)

Quantifies losses from gas dissolution in the electrolyte and parasitic reactions:

$$\eta_F = \frac{V_{\text{meas}}}{V_{\text{theor}}} \quad (34)$$

At steady state with saturated electrolyte, $\eta_F \rightarrow 0.90\text{--}0.95$ for graphite electrodes. During the first 2–3 minutes, $\eta_F < 1$ as gas dissolves in the unsaturated solution.

6.2 Voltage Efficiency (HHV Basis)

$$\eta_V = \frac{V_{\text{th}}}{V_{\text{cell}}} \quad (35)$$

For our case study: $\eta_V = 1.48/5.8 \approx 26\%$ – low, because battery-supplied voltage greatly exceeds V_{th} . Most of the excess drives overpotentials and ohmic drops. This is expected and desirable as a teaching point.

6.3 Overall Energy Efficiency

$$\eta_{\text{en}} = \eta_F \cdot \eta_V \quad (36)$$

6.4 Diagnostic Equivalent Resistance

$$R_{\text{eq}} = \frac{V_{\text{cell}} - V_{\text{rev}}}{I} \quad (37)$$

Useful to quantify total irreversible losses and track electrode degradation over time.

6.5 Ideal Specific Energy Consumption

Theoretical minimum (HHV basis):

$$E_{\text{ideal}} = \frac{\Delta H^\circ}{M_{\text{H}_2}} = \frac{285\,835 \frac{\text{J}}{\text{mol}}}{2.016 \times 10^{-3} \frac{\text{kg}}{\text{mol}}} \approx 39.4 \frac{\text{kW h}}{\text{kg}} \quad (38)$$

Commercial AWE/PEM systems achieve 50–60 kWh kg⁻¹ [8].

7 Thermal Management

7.1 Heat Generation Rate

When $V_{\text{cell}} > V_{\text{th}}$, net heat is dissipated:

$$\dot{Q} = I (V_{\text{cell}} - V_{\text{th}}) \quad (39)$$

When $V_{\text{cell}} < V_{\text{th}}$ (SOEC only), the cell absorbs heat.

For our case study ($I = 128 \text{ mA}$, $V_{\text{cell}} = 5.8 \text{ V}$):

$$\dot{Q} = 0.128 \times (5.8 - 1.48) = 0.55 \text{ W} \quad (40)$$

Negligible heating – the PRUTA container and electrolyte act as an adequate heat sink.

7.2 Membrane Temperature Limit (PEM, for Reference)

Nafion membranes degrade above 80–100 °C (side-chain hydrolysis). Laboratory PEM cells use a PID-controlled water circuit. Not relevant for our open-bath setup, which stays near ambient temperature.

8 Gas Crossover: Fick's Law Permeation Model

In membrane-based electrolyzers, H₂ diffuses through the membrane into the O₂ compartment – a critical safety hazard. By Fick's First Law:

$$\dot{n}_{\text{cross}} = \mathcal{P}_{\text{H}_2} \frac{A}{\delta} \Delta p_{\text{H}_2} \quad (41)$$

where $\mathcal{P}_{\text{H}_2}$ is membrane hydrogen permeability, A is active area, δ is membrane thickness, and Δp_{H_2} is transmembrane pressure difference. For Nafion 117 at 80 °C: $\mathcal{P}_{\text{H}_2} \approx 1.8 \times 10^{-14} \frac{\text{mol}}{\text{m s Pa}}$.

The H₂ mole fraction in the O₂ outlet must remain < 2% (half the LEL = 4%). Emergency hardware interlocks must trigger shutdown if exceeded.

In our open-bath setup (no membrane), the H₂ and O₂ are collected separately by inverted syringes – crossover is geometrically impossible.

9 Electrolyte Design: Na₂SO₄ for Neutral Operation

9.1 Why Na₂SO₄?

For classroom demonstrations, a neutral electrolyte is strongly preferred: it eliminates the hazards of concentrated H₂SO₄ (corrosive acid) or KOH (caustic alkali) while still providing adequate ionic conductivity. Na₂SO₄ is ideal because:

- Non-toxic, non-corrosive at working concentrations
- Na⁺ and SO₄²⁻ are *spectator ions*: they do not react at the electrodes
- Compatible with PP and glass apparatus
- Easily obtainable (pharmacy, Amazon)

9.2 Electrode Potential Competition

Standard reduction potentials in the Na₂SO₄ system (SHE reference, 25 °C):

Table 2: Competing half-reactions in Na₂SO₄ solution.

Half-reaction	E° [V vs SHE]	Outcome
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.000	✓ HER (favoured at cathode)
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	-2.714	× Na deposition impossible
$2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$	1.229	✓ OER (favoured at anode)
$2\text{SO}_4^{2-} \rightarrow \text{S}_2\text{O}_8^{2-} + 2\text{e}^-$	2.010	× Not competitive
$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$	1.358	! Competitive – avoid NaCl

9.3 Kohlrausch Law and Optimal Concentration

For strong electrolytes, the Debye–Hückel–Onsager law (Kohlrausch’s law) gives:

$$\Lambda_m = \Lambda_m^\circ - K\sqrt{c} \quad (42)$$

For Na₂SO₄ at 25 °C [6]: $\Lambda_m^\circ = 259.9 \frac{\text{S cm}^2}{\text{mol}}$, $K \approx 59.7 \text{ S cm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$.

Specific conductivity: $\kappa = \Lambda_m \cdot c$

At $c = 0.5 \text{ M}$: $\kappa \approx 0.06 \frac{\text{S}}{\text{cm}}$; at $c = 1.0 \text{ M}$: $\kappa \approx 0.09 \frac{\text{S}}{\text{cm}}$.

Recommended range: 0.5–1.0 M. Solubility limit $\approx 1.37 \text{ M}$ at 20 °C. Above 1.0 M, marginal conductivity gains and risk of salt precipitation.

Hydrated vs. anhydrous Na₂SO₄: Use **anhydrous** (MW = 142.04 g/mol). The decahydrate form (MW = 322.20 g/mol) requires a different mass calculation and is easy to mistake on the label. For 500 mL of 0.5 M solution: weigh $0.5 \times 0.5 \times 142.04 = \mathbf{35.5 \text{ g}}$.

9.4 Local pH Gradients and Indicator Use

At the cathode, OH⁻ is produced (basic environment, pH↑). At the anode, H⁺ is produced (acidic environment, pH↓). The bulk solution remains near-neutral due to mixing.

Adding 2–3 drops of **bromothymol blue (BTB)** indicator makes these gradients visible: blue/violet at the cathode, yellow/orange at the anode. BTB transition range: pH 6.0 (yellow) – pH 7.6 (blue). Alternatives: phenol red, universal indicator.

10 Electrode Materials

Platinum	Benchmark electrocatalyst for both HER and OER. Lowest η_{act} . Expensive.
Graphite	Chemically inert, low cost, adequate for HER in neutral electrolyte. Higher η_{act} than Pt, but sufficient for demonstration. Pencil leads (graphite + clay binder) work well. Harder grades have more clay and higher electrical resistance (see App. B).
Nickel	Used in AWE cathodes; forms NiOOH/Ni(OH) ₂ OER phase in alkaline. Unsuitable in neutral (moderate corrosion risk).
IrO₂	State-of-the-art OER catalyst for PEM; not available for classroom use.
Avoid	Cu, Fe, Al, Zn: anodic dissolution competes with OER; coloured metallic ions contaminate the solution.

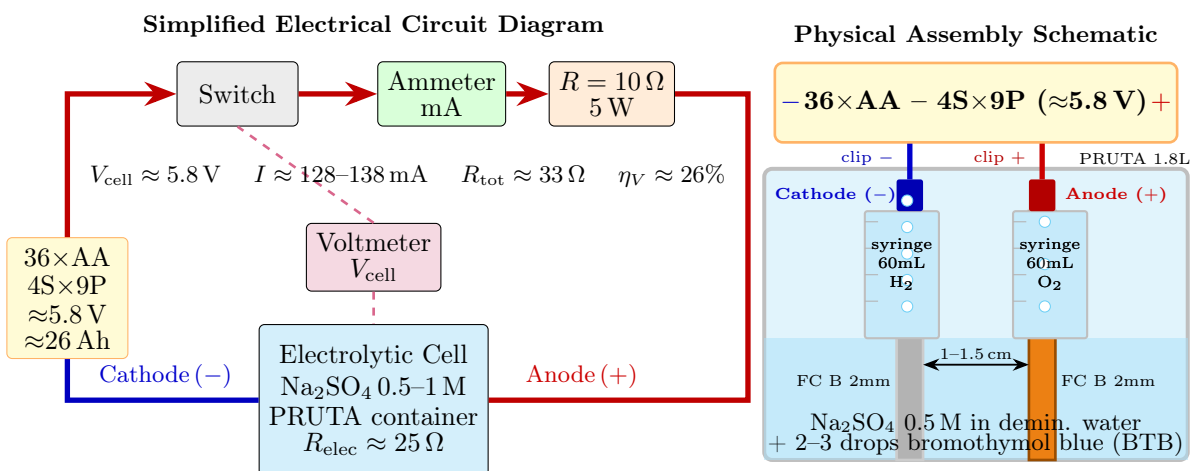
11 Hydrogen Embrittlement

In pressurised systems ($p > p_{\text{atm}}$), atomic hydrogen (produced by HER and then combined at the surface) can diffuse into metallic structural components, reducing ductility and promoting fatigue crack propagation. This phenomenon is known as *hydrogen embrittlement* (HE).

Susceptible materials: high-strength steels ($\sigma_y > 700$ MPa), ferritic alloys, titanium alloys.

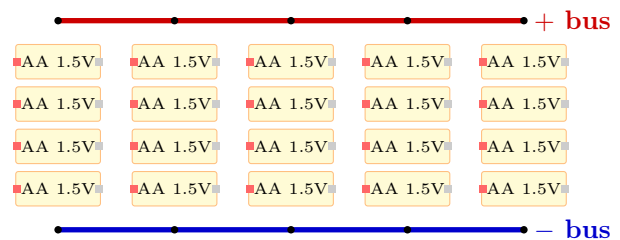
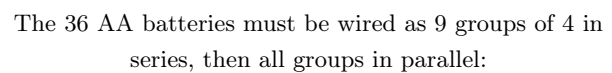
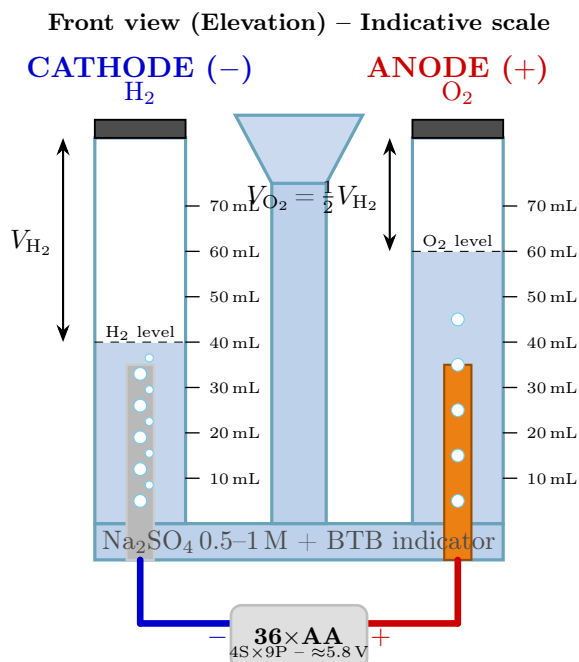
Resistant materials: austenitic stainless steels (AISI 316L – standard choice for H₂ systems), PTFE and Viton[®] for seals. **At near-atmospheric pressure (this Hofmann voltameter setup), HE is not a practical concern.** Borosilicate glass and PP are appropriate.

12 Experimental Setup – Diagrams and Circuits



Note on the 10 Ω / 5 W safety resistor: Not strictly necessary at 128 mA (only 0.16 W dissipated), but recommended because: (a) it limits the initial current surge when the circuit is first closed; (b) it provides a measurable voltage drop $V_R = IR$ that can be used to cross-check the ammeter reading; (c) it protects against accidental short-circuit.

Practical assembly: Use 9 AA battery holders in series-4 configuration (available on Amazon for ≈€1 each). Connect all positive terminals together and all negative terminals together using crocodile-clip jumper wires. The total internal resistance (≈0.22 Ω) is negligible vs. the electrolyte resistance (≈25 Ω).

$$V_{\text{rev}} \approx 1.23 \text{ V} \quad V_{\text{th}} \approx 1.48 \text{ V} \quad V_{\text{cell}} \approx 5.8 \text{ V (AA 4S9P)} \quad I_{\text{est}} \approx 128\text{--}138 \text{ mA}$$


$$\begin{array}{ll} V_{OCV} = 4 \times 1.5 = 6.0 \text{ V} & V_{load} \approx 5.8 \text{ V} \\ C_{tot} = 9 \times 2900 = 26100 \text{ mAh} & R_{int} = \frac{4 \times 0.5}{9} \approx 0.22 \Omega \end{array}$$

13 Case Study – Educational Demonstration

13.1 Materials List with Specifications and Purchase Links

Material	Key parameter	Technical notes	Source
Amazon Basics AA ×36	$V_{\text{nom}} = 1.5 \text{ V}$, $C \approx 2900 \text{ mAh}$	Alkaline, $R_{\text{int}} \approx 0.5 \Omega/\text{cell}$. 4S9P optimal: 5.8 V, 26 Ah. Battery lifetime $\approx 188 \text{ h}$. See Appendix A for AAA vs AA comparison.	Amazon
Faber-Castell TK 9071 2 mm B leads	$d = 2 \text{ mm}$, graphite $\approx 68\%$, clay $\approx 32\%$	$A_{\text{act}} \approx 0.94 \text{ cm}^2$ (30 mm immersed). $R_{\text{el}} \approx 8 \Omega$. Avoid H or 2H: more clay = higher resistance. See Appendix B.	Amazon
Na_2SO_4 Innoxia, anhydrous	Purity $> 99.6\%$, $M = 142.04 \text{ g/mol}$	Weigh 35.5 g for 500 mL of 0.5 M. 1 kg lasts years. Do NOT use decahydrate ($\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$).	Amazon
Demineralised water 5 L	$\rho > 0.1 \text{ M}\Omega \text{ cm}$	Tap water contains Cl^- : avoid (toxic Cl_2 at anode). Distilled/deionised for iron, available at supermarkets.	Amazon
ONFARMA syringes 60 mL ×2	Max 60 mL, PP, graduated every 10 mL	Invert over electrode; extract plunger, seal beak, release. Gas collects at top. Buy in pharmacy ($\approx \text{€}1$ each).	Pharmacy
IKEA PRUTA 1.8 L	$23 \times 16 \times 8 \text{ cm}$, PP	Not hermetic (intentional – open-bath setup). PP is chemically resistant to Na_2SO_4 . Choose 1.8 L box from set.	IKEA
Bromothymol blue (BTB)	pH 6.0 (yellow) to pH 7.6 (blue)	2–3 drops in 500 mL electrolyte. Blue at cathode (OH^-), yellow at anode (H^+). Visible effect after 30 s.	Pharmacy / school supplier
pH strips 1–14 ×240	Universal range	Sample near each electrode with dropper. Records pH more precisely than BTB.	Amazon
Total budget			$\approx \text{€}39$

Table 3: Case study materials, specifications, and purchase sources.

13.2 Battery Configuration: AA 4S × 9P

With 36 Amazon Basics AA batteries in **4 series × 9 parallel**:

$$\begin{aligned}
 V_{\text{OCV}} &= 4 \times 1.5 = 6.0 \text{ V} \quad (\text{open circuit}) \\
 V_{\text{load}} &\approx 4 \times 1.45 = 5.8 \text{ V} \quad (\text{under load, } \approx 0.05 \text{ V/cell drop at } 138 \text{ mA}) \\
 R_{\text{int,tot}} &= \frac{4 \times 0.5 \Omega}{9} \approx 0.22 \Omega \quad (\text{negligible}) \\
 C_{\text{tot}} &= 9 \times 2900 = 26\,100 \text{ mA h} = 26.1 \text{ A h}
 \end{aligned}$$

13.3 Working Current Calculation

Step 1 – Electrode active area:

$$A = \pi \times r \times l_{\text{imm}} = \pi \times 1 \text{ mm} \times 30 \text{ mm} = 94.2 \text{ mm}^2 \approx 0.94 \text{ cm}^2 \quad (43)$$

Step 2 – Electrolyte resistance ($d = 1.5 \text{ cm}$, $c = 1.0 \text{ M}$, $\kappa \approx 0.09 \frac{\text{S}}{\text{cm}}$):

$$R_{\text{elec}} = \frac{d}{\kappa A} = \frac{1.5}{0.09 \times 0.94} \approx 17.7 \Omega \quad (44)$$

(Using 0.5 M: $\kappa \approx 0.06 \text{ S/cm}$, $R_{\text{elec}} \approx 26.6 \Omega$.)

Step 3 – Total resistance:

$$R_{\text{elec}} \approx 25 \Omega \quad (0.5 \text{ M, conservative}) \quad (45)$$

$$R_{\text{leads}} \approx 8 \Omega \quad (\text{two B leads in series}) \quad (46)$$

$$R_{\text{batt}} \approx 0.22 \Omega \quad (47)$$

$$R_{\text{safety}} = 10 \Omega \quad (\text{optional safety resistor}) \quad (48)$$

$$R_{\text{tot}} \approx 25 + 8 + 0.22 = 33 \Omega \quad (\text{without } R_{\text{safety}}) \quad (49)$$

Step 4 – Working current:

$$I = \frac{V_{\text{load}} - V_{\text{rev}}}{R_{\text{tot}}} = \frac{5.8 - 1.23}{33} \approx 138 \text{ mA} \quad (50)$$

With the 10Ω safety resistor: $I \approx \frac{4.57}{43} \approx 106 \text{ mA}$.

Step 5 – Current density:

$$j = \frac{I}{A} = \frac{138 \times 10^{-3}}{0.94 \times 10^{-4}} \approx 1468 \frac{\text{mA}}{\text{cm}^2} \quad (51)$$

This is in the Tafel regime for graphite electrodes – acceptable for a demonstration (graphite does not dissolve, gas evolution is the predominant reaction).

13.4 Electrode Geometry Sensitivity

The table below shows how different electrode diameters and spacings affect I and gas production:

Scenario	d [mm]	A [cm ²]	R_{tot} [Ω]	I [mA]	$\dot{V}_{\text{H}_2}$ [mL/s]	H ₂ in 10 min [mL]
0.5 mm H (Staedtler) + 3 cm spacing	0.5	0.047	535	9	0.001	0.6
2 mm HB + 3 cm spacing	2	0.94	133	34	0.004	2.5
2 mm B + 3 cm spacing	2	0.94	41	111	0.014	8.2
2 mm B + 1.5 cm spacing (recommended)	2	0.94	33	138	0.017	10.2
2 mm B + 1.0 cm spacing	2	0.94	26	176	0.022	13.1
4 mm 4B + 1.5 cm spacing	4	3.77	13	350	0.044	26.2

Table 4: Parametric sensitivity analysis (AA 4S9P, 0.5 M Na₂SO₄, $T = 20^\circ\text{C}$).

Note: R_{leads} estimated from graphite composition; R_{elec} from eq. (23).

Key insight: the dominant parameter is electrode spacing d , not lead hardness.

13.5 Expected Gas Production – Data Table

At $T = 293.15 \text{ K}$ (20°C), $p = 101\,325 \text{ Pa}$, $I = 138 \text{ mA}$ (recommended setup):

$$\dot{V}_{\text{H}_2} = \frac{0.138 \times 8.314 \times 293.15}{2 \times 96485 \times 101325} = 1.72 \times 10^{-8} \frac{\text{m}^3}{\text{s}} = 0.0172 \frac{\text{mL}}{\text{s}} \quad (52)$$

Table 5: Data acquisition table. Grey columns pre-calculated; white columns filled in during experiment. ($I = 138 \text{ mA}$, $T = 20^\circ\text{C}$, $p = 1 \text{ atm}$.)

Δt [s]	I [A]	Q [C]	V_{theor} [mL]	$V_{\text{H}_2}^{\text{meas}}$ [mL]	$V_{\text{O}_2}^{\text{meas}}$ [mL]	η_F	pH cath.
30	0.138	4.1	0.5				
60	0.138	8.3	1.0				
120	0.138	16.6	2.1				
180	0.138	24.8	3.1				
300	0.138	41.4	5.2				
600	0.138	82.8	10.3				
900	0.138	124.2	15.5				

$\eta_F = V_{\text{H}_2}^{\text{meas}}/V_{\text{theor}}$. The first 2–3 min shows lower η_F due to dissolution of gas in unsaturated electrolyte.

13.6 Efficiency and Heat Dissipation

$$\eta_V = \frac{V_{th}}{V_{cell}} = \frac{1.48}{5.8} \approx 25.5\% \quad (53)$$

$$\eta_F \approx 0.90\text{--}0.95 \quad (\text{at steady state}) \quad (54)$$

$$\eta_{en} = \eta_F \cdot \eta_V \approx 23\text{--}24\% \quad (55)$$

$$\dot{Q} = I(V_{cell} - V_{th}) = 0.138 \times (5.8 - 1.48) \approx 0.60 \text{ W} \quad (56)$$

Why is $\eta_V \approx 26\%$ and not higher? The battery voltage (5.8 V) is much higher than the thermoneutral voltage (1.48 V). The surplus 4.3 V is consumed as follows: η_{act} (graphite HER/OER, $\approx 0.5\text{--}0.8$ V each electrode), $\eta_{ohm,elec}$ ($\approx IR_{elec} \approx 3.5$ V), and $\eta_{ohm,leads}$ ($\approx IR_{leads} \approx 1.1$ V). This is entirely expected for a battery-powered demonstration and is an excellent teaching opportunity: ask students to calculate each term and account for the full voltage budget.

13.7 Battery Lifetime

$$t_{life} = \frac{C_{tot}}{I} = \frac{26100 \text{ mAh}}{138 \text{ mA}} \approx \mathbf{189 \text{ hours}} \quad (57)$$

The batteries will run ≈ 189 hours before needing replacement – enough for hundreds of demonstrations. In practice, voltage sag under load will reduce this somewhat, but for this low-drain application (138 mA from 9 parallel groups = 15 mA/group) the batteries operate well within their optimal range.

13.8 Electrolyte Preparation

For $c = 0.5$ M in 500 mL:

$$m = c \times V \times M = 0.5 \times 0.5 \times 142.04 = 35.51 \text{ g} \approx \mathbf{2 \text{ heaped tablespoons}} \quad (58)$$

For $c = 1.0$ M: weigh 71.0 g.

Preparation steps:

1. Weigh 35.5 g of anhydrous Na_2SO_4 (Inoxia) on a kitchen or lab scale.
2. Dissolve in ≈ 400 mL demineralised water in a beaker or jug, stirring.
3. Transfer to the PRUTA container and top up to 500 mL.
4. Add 2–3 drops of BTB indicator. The solution should appear green/yellow-green.
5. For 1.0 M: weigh 71.0 g, same procedure.

Why demineralised water? Tap water typically contains 10–50 mg/L Cl^- . Even at this concentration, some Cl_2 would be evolved at the anode over time, producing an acrid smell and a slight safety hazard. Demineralised water (for steam irons, sold in supermarkets at $\approx \text{€}0.30/\text{L}$) eliminates this problem entirely.

13.9 Why the Staedtler 0.5 mm Leads Are Not Suitable

This is an important negative example. The Staedtler Mars Micro Carbon 250 (hardness H, $\varnothing 0.5$ mm) would give:

$$A = \pi \times 0.25 \text{ mm} \times 30 \text{ mm} \approx 23.6 \text{ mm}^2 \approx 0.24 \text{ cm}^2 \quad (59)$$

$$R_{\text{elec}}(0.5 \text{ mm}, 3 \text{ cm}) = \frac{3}{0.06 \times 0.024} \approx 2083 \, \Omega \quad (60)$$

$$I \approx \frac{5.8 - 1.23}{2083} \approx 2.2 \text{ mA} \quad (61)$$

$$j = \frac{2.2 \text{ mA}}{0.24 \text{ cm}^2} \approx 9 \frac{\text{mA}}{\text{cm}^2} \quad (62)$$

In 10 minutes: $V_{\text{H}_2} \approx 0.16 \text{ mL}$ – invisible. Additionally, the H-hardness composition ($\approx 45\%$ clay) makes the leads mechanically fragile and electrically resistive. The 0.5 mm diameter alone (vs. 2 mm B: A is 16 times smaller) rules out this product.

14 Complete Experimental Procedure

14.1 Before the Experiment

- B1. Check ventilation.** Ensure the room has at least 6 air changes/hour or use a fume hood. Eliminate open flames and spark sources within 1 m.
- B2. Prepare the electrolyte.** Dissolve 35.5 g Na_2SO_4 in 500 mL demineralised water in the PRUTA container. Add 2–3 drops BTB indicator. The solution should be green-yellow.
- B3. Mount the electrodes.** Insert the two FC TK 9071 B leads into an improvised holder (cork, wax board, or 3D-printed clip) with 1.0–1.5 cm electrode separation. The immersed length should be $\approx 30 \text{ mm}$.
- B4. Prepare the syringes.** Extract the plunger from each 60 mL syringe. Seal the Luer-Lock tip with a gloved finger, then invert the syringe over the appropriate electrode. Release the tip: surface tension and atmospheric pressure keep the electrolyte inside. Gas will accumulate at the top.
- B5. Wire the batteries.** Assemble 9 battery holders ($4 \times \text{AA}$ each) in series (check polarity: $+\rightarrow-\rightarrow+\rightarrow-$ within each group). Connect all holders in parallel: all $+$ terminals to the red bus, all $-$ terminals to the blue bus.
- B6. Connect crocodile clips.** Blue to cathode (left lead), red to anode (right lead). Verify polarity before applying power.
- B7. Record ambient conditions.** Measure T ($^\circ\text{C}$) and p (hPa) for later calculation of V_{theor} using eq. (28).

14.2 During the Experiment

- D1. Close the circuit.** Attach the last crocodile clip. Verify gas bubbles appear at both electrodes within 15–30 seconds. If no bubbles: check polarity, connections, and electrode contact.
- D2. Data logging.** Every 30–60 s fill in the table (Section 13): read gas volumes from the syringe graduations, ammeter current, and voltmeter reading.
- D3. pH measurement.** With a micropipette or dropper, collect a drop of solution near each electrode. Apply to a pH strip. Record. You should see $\text{pH} > 9$ near cathode and $\text{pH} < 5$ near anode after 5 minutes.
- D4. Compute η_F .** For each row: $\eta_F = V_{\text{meas}}/V_{\text{theor}}$. Discuss the low initial values (dissolved gas effect) and the approach to steady state.
- D5. Observe:** the 2:1 $\text{H}_2:\text{O}_2$ volume ratio (Faraday’s Law); the colour gradient in the BTB indicator; any electrode discolouration (slight surface oxidation of graphite at anode is normal and reversible).

14.3 After the Experiment

- A1. Power off.** Disconnect the batteries first, then the crocodile clips.
- A2. Vent gases.** Slowly lift the syringes under a fume hood or near an open window. The small volumes (10–15 mL H₂) dilute immediately in room air and pose no hazard.
- A3. Dispose of solution.** Na₂SO₄ at 0.5 M is non-hazardous; drain down the sink with copious water. Check local regulations.
- A4. Rinse leads.** Rinse FC leads with demineralised water, pat dry. Inspect for fractures (rare with B hardness at these current densities).
- A5. Post-processing.** Plot V_{meas} vs. Δt , overlay V_{theor} , compute η_F at each point, and discuss efficiency losses.

15 Safety Regulations

Hydrogen hazard profile:

Lower Explosive Limit (LEL) = 4.0 % v/v in air
 Upper Explosive Limit (UEL) = 75.0 % v/v in air
 Minimum ignition energy $\approx$ 0.02 mJ (lowest of any flammable gas)
 Autoignition temperature: 500–571 °C
 Density at ambient: 0.0837 kg/m³ (buoyant – accumulates at ceiling)

At 138 mA, H₂ production rate $\approx$ 0.017 mL/s – dilutes rapidly in well-ventilated spaces.

ISO 15916:2015 Basic considerations for the safety of hydrogen systems. Covers fundamental hazards and risk-reduction principles.

ISO 22734:2019 Hydrogen generators using water electrolysis. Construction, safety and performance requirements.

IEC 60079-10-1 Explosive atmospheres – ATEX zone classification for areas with flammable gases.

CEI EN 60079-10-1 Italian transposition of IEC 60079-10-1.

ATEX 2014/34/EU EU directive on equipment in potentially explosive atmospheres.

Mandatory laboratory measures:

- Adequate ventilation: minimum 6 air changes/hour or fume hood.
- No open flames, sparks, or hot surfaces within 1 m.
- Two-person rule: never conduct hydrogen experiments alone.
- CO₂ or dry powder extinguisher within reach.
- Never mix H₂ and O₂ from the syringes before venting separately.
- Inspect glassware/syringes for cracks before use.

Perspective on risk: At 138 mA, the H₂ production rate is $\approx 0.017 \text{ mL/s} = 1 \text{ L/h}$. A well-ventilated room of 50 m³ (6 air changes/hour) has an air flow of 5000 L/min. The H₂ concentration at steady state is $\approx 1 \text{ L/h} / (5000 \text{ L/min} \times 60) \approx 3 \times 10^{-6}$, well below any detectable level. For a classroom demonstration of this scale, good natural ventilation is sufficient.

16 Summary of Key Equations

Table 6: Complete reference of the key equations.

Quantity	Equation	Eq.
Global reaction	$\text{H}_2\text{O}(l) \rightarrow \text{H}_2(g) + \frac{1}{2}\text{O}_2(g)$	(1)
Gibbs–Helmholtz	$\Delta H = \Delta G + T\Delta S$	(8)
Reversible voltage	$V_{\text{rev}} = \Delta G/(zF) = 1.23 \text{ V}$	(12)
Thermoneutral voltage	$V_{\text{th}} = \Delta H/(zF) = 1.48 \text{ V}$	(14)
Nernst equation	$V = V_{\text{rev}} + \frac{RT}{zF} \ln Q_r$	(16)
Temperature dependence	$dV_{\text{rev}}/dT \approx -8.5 \times 10^{-4} \text{ V/K}$	(17)
Real cell voltage	$V_{\text{cell}} = V_{\text{rev}} + \eta_{\text{act}} + \eta_{\text{ohm}} + \eta_{\text{conc}}$	(18)
Butler–Volmer	$j = j_0[\exp(\alpha_a F \eta / RT) - \exp(-\alpha_c F \eta / RT)]$	(19)
Tafel equation	$\eta_{\text{act}} = (RT/\alpha F) \ln(j/j_0)$	(20)
Ohmic overpotential	$\eta_{\text{ohm}} = IR_{\text{eq}}, R_{\text{eq}} = d/(\kappa A)$	(22), (23)
Concentration overpotential	$\eta_{\text{conc}} = (RT/zF) \ln(1 - j/j_L)$	(24)
Faraday – moles H_2	$n_{\text{H}_2} = I\Delta t/(2F)$	(26)
Volume H_2	$V_{\text{H}_2} = I\Delta t RT/(2Fp)$	(28)
Volumetric flow rate	$\dot{V}_{\text{H}_2} = IRT/(2Fp)$	(29)
Faradaic efficiency	$\eta_F = V_{\text{meas}}/V_{\text{theor}}$	(34)
Voltage efficiency	$\eta_V = V_{\text{th}}/V_{\text{cell}}$	(35)
Overall efficiency	$\eta_{\text{en}} = \eta_F \cdot \eta_V$	(36)
Diagnostic resistance	$R_{\text{eq}} = (V_{\text{cell}} - V_{\text{rev}})/I$	(37)
Heat generated	$\dot{Q} = I(V_{\text{cell}} - V_{\text{th}})$	(39)
Gas crossover (Fick)	$\dot{n}_{\text{cross}} = \mathcal{P}(A/\delta)\Delta p$	(41)
Kohlrausch law	$\Lambda_m = \Lambda_m^\circ - K\sqrt{c}$	(42)

A Appendix A – AA vs AAA Battery Comparison

Parameter	AA (Amazon Basics)	AAA (Amazon Basics)	Notes
Nominal voltage	1.5 V	1.5 V	identical
Typical capacity	≈ 2900 mAh	≈ 1200 mAh	AA $\times 2.4$
Internal resistance	$\approx 0.5 \Omega$ /cell	$\approx 1.0 \Omega$ /cell	AA half
V_{OCV} (4S9P)	6.0 V	6.0 V	identical
V_{load} (4S9P, 138 mA)	≈ 5.82 V	≈ 5.74 V	diff. 1.4%
C_{tot} (4S9P)	26.1 Ah	10.8 Ah	AA $\times 2.4$
$R_{\text{int,tot}}$ (4S9P)	$\approx 0.22 \Omega$	$\approx 0.44 \Omega$	negligible both
Battery lifetime at 138 mA	≈ 189 h	≈ 78 h	both: excess
Estimated I (same $R_{\text{tot}} = 33 \Omega$)	≈ 139 mA	≈ 137 mA	diff. $< 2\%$
H_2 in 10 min	≈ 10.3 mL	≈ 10.2 mL	practically identical
Cost 36 units (Amazon.it)	$\approx \text{€}15$	$\approx \text{€}10$	AAA cheaper

Table 7: Both $36 \times \text{AA}$ and $36 \times \text{AAA}$ work for this experiment. Comparison between $4\text{S} \times 9\text{P}$. The electrical performance is virtually identical (current difference $< 2\%$). AA batteries provide $2.4 \times$ longer runtime, which is academic given that either configuration lasts tens of hours. Choose based on price and availability.

B Appendix B – Lead Hardness and Diameter

Grade	Graphite [%]	Clay [%]	Resistivity [relative]	R_{el} [est. Ω]	I (AA 4S9P) [mA]	$\text{H}_2/10$ min [mL]
4B	≈ 85	≈ 15	very low	≈ 4	155	11.5
2B	≈ 75	≈ 25	low	≈ 6	150	11.1
B	≈ 68	≈ 32	medium	≈ 8	138	10.2
HB	≈ 60	≈ 40	medium-high	≈ 13	120	8.9
H	≈ 55	≈ 45	high	≈ 18	100	7.4
2H	≈ 45	≈ 55	very high	≈ 28	72	5.3

Table 8: Graphite composition and electrical properties by pencil lead hardness. Estimates for FC TK 9071 2 mm leads; $d = 1.5$ cm; $0.5 \text{ M Na}_2\text{SO}_4$; $R_{\text{elec}} \approx 25 \Omega$. All grades produce visible bubbles.

Lead type	$\varnothing$ [mm]	A_{act} [cm ²]	R_{elec} [Ω]	I [mA]	$\text{H}_2/10$ min [mL]
Staedtler 0.5 mm H (micro)	0.5	0.047	532	9	0.7
Mechanical pencil lead 0.7 mm	0.7	0.066	378	13	1.0
Mechanical pencil lead 1.0 mm	1.0	0.094	265	17	1.3
FC TK 9071 2 mm B	2.0	0.94	25	138	10.2
Thick pencil lead 4 mm 2B	4.0	3.77	6.2	300	22.2
Carpenter pencil $\approx 5 \times 2$ mm	~ 5	9.4	2.5	420	31.1

Table 9: Effect of lead diameter on active area and performance (hardness B, $d = 1.5$ cm, AA 4S9P, $0.5 \text{ M Na}_2\text{SO}_4$). The 2 mm diameter provides a $20 \times$ larger active area than 0.5 mm leads.

C Appendix C – Budget Summary

Item	Source	Quantity	Cost
Amazon Basics AA batteries ×36	Amazon	36	≈€15
Faber-Castell TK 9071 2 mm B leads	Amazon	1 pack	≈€4
Na ₂ SO ₄ anidro Innoxia 1 kg	Amazon	1 kg	€8.59
Demineralised water	Amazon/Supermarket	5 L	€1.19
ONFARMA syringes 60 mL	Pharmacy	2	≈€2
IKEA PRUTA container 1.8 L	IKEA	1 set	€4.95
Bromothymol blue (BTB) indicator	Pharmacy / school supplier	1 bottle	≈€3
pH strips 1–14 (×240)	Amazon	1 pack	≈€4
Battery holders 4×AA ×9 (optional)	Amazon	9	≈€9
Safety resistor 10 Ω / 5 W (optional)	Electronics shop	1	≈€0.50
Total (without optional)			≈€39
Total (with optional)			≈€48

Table 10: Complete materials budget.

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